

SuperMoTo: an integrated monitoring matrix, room correction and Ambisonics decoding plugin for audio production and immersive listening rooms

Olivier Doaré

ENSTA Paris, Institut Polytechnique de Paris

`olivier.doare@ensta.fr`

Abstract

SuperMoTo is an open-source audio plugin (VST3, AU and Standalone) that puts a studio’s monitoring signal chain into one plugin instance. From that instance it routes up to 32 inputs to 32 outputs through a configurable gain-and-EQ matrix, applies per-loudspeaker FIR correction filters derived from in-room acoustic measurements, time-aligns the loudspeakers of a rig and phase-aligns a subwoofer to the corrected main speakers, and generates periphonic Ambisonics decoders up to third order for immersive listening rooms. Acoustic measurement tools and a transfer-function analysis and correction design engine are built in, so the whole calibration workflow happens inside the plugin, one loudspeaker at a time or for a whole rig in a single pass. It is implemented in C++ using the JUCE framework and is licensed under the AGPL 3.0, or under commercial terms for holders of a commercial JUCE licence.

1 Introduction

Decisions about balance, timbre and spatial imagery are all made at the listening position, so a systematic error in what the engineer hears accumulates through the whole mix. In a real room, what the engineer hears is never the loudspeaker alone but the combined response of the loudspeaker, the room, and any bass-management or routing in the signal path. Three independent classes of error contribute to this. Room modes below the Schroeder frequency produce large, position-dependent peaks and dips with audible ringing, speaker-boundary interference and early reflections add comb-filtering through the midrange, and the loudspeaker’s own magnitude and phase response is rarely flat [15]. Electronic correction cannot cure all of these (room cancellation notches and late reflections are not minimum-phase and cannot be inverted by any causal filter [4]), but the broad, reproducible, minimum-phase portion of the combined loudspeaker-room response can be usefully attenuated with a well-designed FIR filter, which reduces coloration and improves translation to other playback systems.

No single open-source tool covers this workflow end to end. The widely-used Room EQ Wizard (REW)¹ provides acoustic measurement and filter design as a standalone desktop application but not as a DAW plugin, requiring the resulting correction impulse responses to be loaded into a separate convolution plugin and routing to be managed outside the tool. Commercial room correction products such as Sonarworks SoundID Reference² and IK Multimedia ARC³ focus exclusively on correction and do not integrate routing, bass management, or immersive audio decoding. High-channel routing tools and Ambisonics decoders exist as separate plugins, the IEM Plug-in Suite⁴ being the reference open-source example, but are not combined with a calibration workflow. SuperMoTo covers the whole workflow in one plugin instance, namely the monitoring matrix, the measurement engine, the correction design, the FIR convolution and the Ambisonics decoder generation. It can also read the decoding matrices exported by IEM’s AllRADecoder, so an all-round decoder can be driven by a set of measured, individually corrected loudspeakers. The combination matters most in VR and immersive audio production, where a loudspeaker array has to

¹<https://www.roomeqwizard.com/>

²<https://www.sonarworks.com/soundid-reference>

³<https://www.ikmultimedia.com/products/arcstudio/>

⁴<https://plugins.iem.at/>, GPLv3, from the Institute of Electronic Music and Acoustics, Graz.

be routed, time-aligned, individually corrected and driven by a decoded Ambisonics signal at the same time.

Aligning a subwoofer’s phase to the mains is itself established practice rather than a new idea, and Section 7 places the construction used here against the parametric all-pass designs it descends from. The contributions of this paper are the following.

- A monitoring chain that carries routing, per-loudspeaker EQ, time alignment, FIR correction and Ambisonics decoding in one plugin instance, with a latency budget that keeps a rig aligned exactly as the user set it for the least added latency (Section 3).
- A closed-form subwoofer phase alignment, read directly off the measured phase and carried by the correction filter that is already there, in place of a parametric all-pass fitted by an optimiser, as done by other solutions (Section 7).
- A power-weighted crossover-band group-delay estimator for the main-to-subwoofer offset, and the identity that makes one such offset per loudspeaker correct across a whole rig sharing a single subwoofer (Sections 7 and 8).
- An all-in-one open-source implementation in which the measurement, the analysis, the filter design and the playback chain are the same instance, so a rig can be calibrated without leaving it (Section 10).

2 The monitoring chain

The routing matrix connects up to 32 inputs to 32 outputs through a grid of crosspoint *frames*. Six independent configurations (A–F) are stored per preset and can be engaged in exclusive mode (instant A/B switching between monitoring rigs) or summed. Everything the plugin does to the audio happens in one pass over the block, in the order the signal physically travels. With X_i the inputs and $\omega = 2\pi f$, the signal reaching output o is

$$Y_o(\omega) = g \underbrace{e^{-j\omega k_o/f_s}}_{\text{latency comp.}} \underbrace{\tau_o E_o(\omega) e^{-j\omega \delta_o/f_s} C_o(\omega)}_{\text{loudspeaker } o} \sum_{c \in \mathcal{A}} \sum_i \underbrace{\gamma_{cio} B_{cio}(\omega)}_{\text{crosspoint}} X_i(\omega), \quad (1)$$

where $\mathcal{A}$ is the set of engaged configurations, k_o and δ_o are delays in samples at the sampling rate f_s , and g is the master gain (level, mute and dim on one smoother).

A crosspoint contributes a gain and polarity γ_{cio} , ramped linearly over 50 ms so that switching a route, or a whole configuration, cannot click, followed by its own two-band EQ B_{cio} , which is where a configuration-specific tweak on that one route belongs. The per-output terms are the trim and polarity τ_o , a second two-band EQ E_o , a delay of δ_o samples (up to 100 ms) for physical speaker time-alignment, and the FIR correction C_o , applied by a zero-latency partitioned convolution engine [16]. Each EQ band is a peaking biquad, or a lowpass, highpass or bandpass of second or fourth order, a fourth-order section being two biquads carrying the Butterworth Q pair scaled by the band’s own Q setting. The delay is rounded to a whole number of samples and read from a ring buffer at one tap, so that it is a pure sample shift and the same operation on every output. The half sample given up is 11 μ s at 44.1 kHz, 4 mm of path, and one sample of relative error between two outputs places its first cancellation notch at $f_s/2$, so it cannot comb inside the band. Interpolating between two taps instead, as this did until the measurements of Section 9 showed it up, colours the treble by where the delay falls between samples — 1.5 dB at 8 kHz and 7.6 dB at 16 kHz for half a sample, and differently on each output. A windowed-sinc interpolator removes that colouration at about ten times the arithmetic and is kept in the source for a rig that needs the resolution; for a listening room, rounding removes it at none.

Everything that describes a physical loudspeaker therefore multiplies once, after the sum, and is shared by every configuration (A...F) that drives it, while the routing and its per-route tweaks belong to one configuration. When filters of different lengths are loaded on different outputs, automatic inter-output latency compensation keeps every speaker time-aligned at the listening position for the least added latency (Section 3).

A configuration tool generates standard stereo and surround layouts (2.0 through 7.1) with bass management and writes a complete matrix configuration ready for refinement. The subwoofer is fed either from a dedicated input (an LFE channel) or from the sum of the main channels, and bass management is written as a highpass on each main output and a lowpass on the subwoofer’s.

3 Latency and time alignment

The partitioned convolution engine adds no block latency whatever the filter length, but a linear-phase correction is centred on its own impulse, so output o carries a bulk latency ℓ_o equal to the position of the peak of its loaded impulse response (about half the filter length, and near zero for a minimum-phase filter). Filters of different lengths on different outputs would therefore introduce timing inconsistencies between them. Writing d_o for the delay the user (or automatic analysis) set on output o in samples and $(\cdot)_+$ for the positive part, the delay line actually runs only the remainder $\delta_o = (d_o - \ell_o)_+$, the uniform compensation is

$$k = \max_{o \text{ fed}} (\ell_o - d_o)_+, \quad (2)$$

and each fed output takes the share $k_o = d_o + k - \max(d_o, \ell_o)$ of it, so that

$$\delta_o + \ell_o + k_o = d_o + k \quad \text{on every fed output.} \quad (3)$$

The rig is aligned exactly as wanted, offset by a single common term k , which is the smallest value satisfying $d_o + k \geq \ell_o$ everywhere and is zero whenever each output's own alignment delay already covers its own filter, the usual case for a rig that has been time-aligned. An output that no engaged configuration feeds is left out of the maximum, so an unused loudspeaker's long filter cannot impose latency on the rig.

4 Ambisonics decoding

The configuration tool also generates periphonic (3D) Ambisonics sampling decoders for orders 1 through 3, following the AmbiX convention (ACN channel order, SN3D normalisation) [18]. For each loudspeaker at azimuth θ , elevation φ and radius r , the decode gain is

$$D[s][j] \propto a_{n(j)} Y_j^{\text{SN3D}}(\Omega_s), \quad (4)$$

where a_n are the $\max\text{-}r_E$ per-degree weights derived from the largest root of the Legendre polynomial P_{N+1} [19], which improves high-frequency localisation stability. The rig is described by a loudspeaker count and a table of azimuth, elevation, radius and output channel, starting either from a hand-tuned regular layout for the order's canonical count or from a near-uniform spherical Fibonacci lattice for any other count. Non-equidistant speaker rigs are compensated by per-speaker delay and trim derived from the inverse-square law, each of which can be left out so that an alignment or a level already obtained from measurements survives a later re-application of the decoder. Bass management low-passes the omnidirectional W component to a subwoofer and high-passes the speaker feeds.

Irregular and hemispherical rigs are better served by an all-round decoder than by a plain sampling decoder, so the tool can instead import the `.json` configuration exported by the IEM AllRADecoder and use its matrix, converting it to the plugin's SN3D/ACN feed and applying the same radius compensation.

5 Acoustic measurement

The measurement view drives the plugin outputs with a band-limited stimulus (5s to 30s duration) and records the microphone response on one of the plugin's input channels. The sweep stimulus is a *synchronized* exponential sweep in the sense of [2], its sweep rate being quantised so that the stimulus phase at each harmonic arrival time is an exact multiple of 2π , which is what allows the distortion orders to be separated with their true phase at analysis time (Section 6). White noise is offered as the alternative stimulus.

Each captured stereo file stores the raw stimulus on channel 1 and the microphone on channel 2, so the transfer function estimate at analysis time is always the ratio of the captured microphone signal to the sent stimulus. Three modes are provided, "Dry" (bare loudspeaker and room, correction bypassed), "FIR" (single output with its correction in circuit, for before and after verification), and "System" (complete engine including routing and all FIR filters, for end-to-end validation).

A run measures every selected channel in turn with the microphone in one place and writes its files into a measurement folder that numbers positions per channel automatically and tags an optional subwoofer channel. The folder carries a generated XML manifest recording, for every run, its time, its stimulus parameters, a free-text comment, the microphone calibration in effect (name and a verbatim copy of

the file) and the dB SPL calibration if one has been made by reading a real sound level meter. The folder is therefore self-describing, and the group-alignment mode of Section 8 loads one in a single step. Microphone frequency-response calibration files in the REW/miniDSP plain-text format are supported and are applied at analysis time rather than baked into the recordings, so the calibration can be updated, or replaced by the copy embedded in a folder measured years earlier, without re-measuring.

6 Transfer-function estimation and correction design

For each set of measurements (one speaker, multiple microphone positions), the plugin offers two estimators for the loudspeaker-room transfer function. The first is the Welch H_1 estimator [1]

$$\hat{H}(f) = \frac{\hat{P}_{xy}(f)}{\hat{P}_{xx}(f)}, \quad (5)$$

with Hann-windowed 50 %-overlapped segments and a user-selectable window length of 16 384 to 262 144 samples, and it applies to any stimulus. The second, available and selected by default when the run’s manifest identifies it as a synchronized sweep, deconvolves the recording by the analytic spectral inverse of the sweep [2], which yields the full-band response from a single sweep and requires no regularised division by a measured spectrum. Because a distortion product of order m is emitted $L \ln m$ seconds earlier than the linear response at the same frequency (L being the sweep rate), the deconvolved response separates the orders along the time axis. The linear impulse response is windowed out for the estimate, and orders 2 to 5 are retained as distortion curves, power-averaged across positions. Under the Welch estimator those same products are instead folded into the linear estimate. The two methods share the entire downstream chain, so a set can be re-analysed either way for comparison.

Before averaging across positions, the bulk propagation delay of each measurement is estimated from the impulse response peak and removed, so position-dependent phase ramps do not cancel in the spatial average. The resulting complex spatial average follows the practice of correcting the broad trend common to the listening area rather than a single-seat interference pattern [15, 3].

The correction filter is the regularised complex inverse of the frequency-dependent-smoothed average. A variable fractional-octave smoothing is applied, where the octave fraction interpolates log-linearly from an independent low-frequency setting to a high-frequency setting, allowing the correction to track room modes finely at low frequencies while responding only to broad trends above a few kilohertz [5]. Boosts are limited by a smooth tanh soft-knee ceiling to avoid ringing and headroom loss at room cancellation notches [6], and the correction fades back to unity outside a user-set analysis range, where neither the loudspeaker nor the measurement carries usable information.

For subwoofer-equipped systems, a second set of measurements (the sub at the same microphone positions) is loaded, and the phase of the main’s correction is steered onto the subwoofer’s across the crossover region by an all-pass factor, so that the two sum coherently there. That design is developed in Section 7.

The final correction spectrum is exported as a mono 32-bit float impulse-response wav file in either *linear-phase* (energy centred at $N/2$, enabling full phase and subwoofer alignment at the cost of $N/2$ samples of latency) or *minimum-phase* (real-cepstrum method, magnitude-only correction with essentially zero bulk latency, suited to tracking sessions) form [17].

7 Subwoofer phase alignment

A main loudspeaker and a subwoofer only sum to a flat response through the crossover when they are in phase there [7]. A fourth-order Linkwitz-Riley lowpass contributes a group delay of $2\sqrt{2}/\omega_x$ at DC and at its corner, 5.6 ms at $f_x = 80$ Hz, and the low-frequency acoustic centre of a loudspeaker moves with frequency as well [14]. Several milliseconds of the offset are therefore built into the system, and no repositioning of the box removes them.

Steering a main’s phase onto the subwoofer’s through the crossover has been proposed by a few authors or constructors. Audyssey’s phase equalisation cascades second-order all-pass sections onto the satellite channel and computes their pole angles and radii recursively, so as to minimise $\varphi_{\text{sub}} - \varphi_{\text{sat}} - \varphi_{\text{AP}}$ over a weighted band around the crossover [9, 10], and Dirac Live Bass Control applies a pre-specified number of all-pass biquads to the high- and low-frequency branches so that the subwoofer and the channel are

in phase around the crossover at a chosen position in the room [11]. The crossover region remains an active subject [12]. In each of these the all-pass is a parametric filter of chosen order whose coefficients are found by an optimiser against a phase-error criterion. What follows shares that objective and differs in construction. The alignment is read straight off the measured phase in closed form, on the same frequency grid the correction is designed on, so there is no filter order to choose, no objective function and no convergence to check, and it is carried by the FIR that already applies the magnitude correction, so it costs nothing at run time.

Aligning the two first requires their relative timing to survive the analysis, which by default it does not. The bulk propagation delay of each measurement is estimated from its impulse-response peak and removed before the positions are averaged, since otherwise the position-dependent phase ramps cancel in the average. Each subwoofer capture is instead delay-anchored on the main capture taken at the same position, shifted by the main's delay rather than by its own, so the averaged subwoofer response $\bar{S}$ keeps its true phase relative to the main's average.

The alignment itself is a single factor applied to the correction spectrum. From the subwoofer's unit-magnitude phasor $\hat{s} = \hat{S}/|\hat{S}|$, where $\hat{S}(f) = \pm \bar{S}(f) e^{j2\pi fT}$ carries the subwoofer polarity and an assumed bulk delay T discussed below, the correction C designed for the main is multiplied by

$$A(f) = \frac{(1 - w_x(f)) + w_x(f) \hat{s}(f)}{|(1 - w_x(f)) + w_x(f) \hat{s}(f)|}, \quad (6)$$

where the alignment weight is a raised cosine in log frequency,

$$w_x(f) = \frac{1}{2} + \frac{1}{2} \cos(\pi x(f)), \quad x(f) = \min\left(1, \max\left(0, \frac{1}{N_{\text{oct}}} \log_2 \frac{f}{f_x}\right)\right), \quad (7)$$

so $w_x = 1$ at and below the crossover f_x and $w_x = 0$ at and above $2^{N_{\text{oct}}} f_x$, with a release width N_{oct} of one octave by default. Above the release the main dominates and keeps the phase its own correction gives it.

Three properties of (6) are what make it usable. Taking the phasor discards the subwoofer's magnitude, so none of the subwoofer's own response leaks into the main's correction. The renormalisation makes $|A| = 1$ at every bin, so the alignment is strictly a phase operation and leaves the designed magnitude correction untouched, which is what allows the same design to serve with and without a subwoofer. And the blend is taken on the phasors rather than on an unwrapped phase angle, so the interpolation travels the short arc between 1 and $\hat{s}$ and the rotation A applies is bounded by π at every frequency, by construction and with no phase-unwrapping step, which would need an anchor in a band where the subwoofer is often below the noise. Scaling an unwrapped angle, anchored at the crossover, turns out to be equivalent once the bulk delay below has been removed, and worse when it has not.

The delay T is what keeps the all-pass small. Advancing the subwoofer phasor by T before its phase is taken leaves the all-pass carrying only the residual, while the bulk of the offset is applied as an ordinary delay on the main's output, where it costs nothing. T is a relative quantity, not the delay applied to the main. If the outputs receive bulk delays a_m and a_s , the phase difference left between the two acoustic paths through the crossover region is $2\pi f [T - (a_m - a_s)]$, which vanishes at every frequency only if

$$T = a_m - a_s. \quad (8)$$

The group-alignment mode designs each speaker's correction with exactly its own $a_i - a_s$, so (8) holds for every speaker of the rig and for any trim applied to the subwoofer against the group.

T is estimated from the measurements themselves, and the estimator has to be chosen with care, because the two available measures of lateness are not measuring the same quantity. An arrival time read from an impulse-response peak is accurate to about the inverse of the measured bandwidth [13], some 50 μs for a full-range loudspeaker but 10 ms for a subwoofer that passes nothing above 100 Hz, so it is weakest exactly where it is wanted. The plugin instead fits the slope of the anchored subwoofer's unwrapped phase over $[f_x/2, 2f_x]$ by least squares weighted by the measured power $|\bar{S}|^2$, which returns the effective group delay of the subwoofer in the crossover region, geometry and filtering together, and that is what decides whether the two sources add or cancel. Weighting each point of that fit by $|\bar{S}|^2$ lets the part of the band where the subwoofer actually has output decide the slope, which is what makes the estimate stable.

Because A carries phase, the correction is a mixed-phase filter and is not causal on its own, which is why the exported impulse response is centred at $N/2$ and why it brings the bulk latency the compensation of Section 3 then absorbs. A minimum-phase export rebuilds the filter from $|C|$ alone and A has unit magnitude, so the subwoofer alignment disappears from it entirely. The loudspeakers still arrive together in that mode, since the delays are written to the outputs regardless, but there is no phase steering through the crossover.

8 Multi-speaker group alignment

For rigs of more than a couple of loudspeakers, a group-analysis mode holds one transfer-function estimate per speaker (up to 16, plus one shared subwoofer when the rig has one) side by side, sharing a single set of correction-design settings across all of them. A whole measurement folder is loaded in one step, the manifest supplying which channels were measured, which one is the subwoofer and which output each speaker sits on. Because every channel of a run was measured with the microphone in one place, each subwoofer capture is paired with the speaker capture of the same run, which keeps the pairing correct even when a speaker missed a run. Individual runs can be excluded from the analysis, and runs measured through an existing correction are excluded by default, since a correction has to be designed from the uncorrected loudspeaker.

Each speaker’s bulk propagation delay, estimated as in Section 6, is compared across the group and every driver is delayed to match the most distant one, so the whole rig arrives coherently at the reference position. The resulting delay is fed back into each speaker’s own subwoofer phase-alignment term before its correction filter is finalised. A level-matching trim per speaker is also derived, as the mean power of its corrected response over the 500 Hz to 2000 Hz band that SMPTE ST 2095-1 specifies for channel level calibration [8], relative to the quietest speaker of the group. It is reported rather than applied, so that a deliberate level offset is never overwritten.

The subwoofer itself is never given a correction FIR filter. It contributes only its computed alignment delay and polarity (plus eventual gain lowpass order 2/4 crossover filter as given by the config tool or tuned manually in the matrix).

Applying the result exports every speaker’s correction impulse response, writes the delays, the filters and the subwoofer’s polarity onto the corresponding outputs, and saves a markdown report listing the settings, the files used, the measured and applied delays, the suggested trims and, optionally, the plotted responses. Loading, re-analysing and exporting a whole group run on a background thread pool with progress feedback, so the editor stays responsive with long, multi-position measurement sets across many speakers, and a batch already running survives the editor being closed.

9 Validation

The alignment was evaluated on a two-way stereo rig with one subwoofer in a treated studio, crossed over at 85 Hz. The loudspeakers and the reference position form a triangle with 1.70 m sides and a 2.55 m base, so the pair is wide and subtends nearly 100° , and the subwoofer stands on the floor between them, 1.6 m from the reference position. Each main and the subwoofer were first measured on their own, with the correction bypassed, at ten microphone positions over the listening area: six spanning 45 cm along the axis and 34 cm across it, and four of them repeated 8 cm higher, each with one 15 s synchronized sweep per channel. One group analysis was run on those captures and exported twice, linear phase and minimum phase, one minute apart with nothing else touched. The system was then measured through the matrix at the same ten positions in three states, visited one after another at each position without moving the microphone: with no correction, with the linear-phase correction, and with the minimum-phase one. The three states are three configurations of the matrix holding the same delays, the same bass-management filters and the same subwoofer feed. That last point deserves a check rather than an assumption, because each configuration routes into the subwoofer through its own crosspoint and the analysis report states the polarity of the subwoofer *output*, not of the route into it: fitting each state’s polarity against the separate captures puts all three on the same sign, at 0.90 dB to 0.95 dB RMS where the opposite sign costs 2.7 dB to 5.7 dB.

Comparing several alignment strategies by measurement alone would mean loading each one into the rig and re-measuring every position, once per strategy, with the room, the loudspeakers and the exact

microphone placement free to drift from one session to the next. Each strategy is instead computed from a single set of recordings, the one in which each main and the subwoofer were measured on their own, and the summation it would produce is predicted from them. Every strategy is then scored on exactly the same data, and whatever separates two of them comes from the strategy rather than from the session. Because each subwoofer capture is delay-anchored on the main capture of the same position, the same propagation delay d_p has been removed from both, and the acoustic sum at position p is exactly

$$Y_p(\omega) = e^{-j\omega T} H_{hp}(\omega) C(\omega) H_p(\omega) \pm H_{lp}(\omega) S_p(\omega) \quad (9)$$

up to the common factor $e^{-j\omega d_p}$ that the anchoring took off both terms, which is a pure delay and changes no magnitude. Here H_p and S_p are the measured responses of the main and of the subwoofer at that position, and H_{hp} , H_{lp} the bass-management filters. Nothing is simulated except the filters the plugin would itself apply. Every number reported below comes from a leave-one-position-out design. To score position p , the correction is derived from the nine other positions only and then evaluated at p , and this is repeated with each of the ten positions held out in turn, so no filter is ever scored on a measurement that helped shape it.

The run with no correction turns the check of Figure 1(a) into a direct one. Dividing the corrected measurement by the uncorrected one taken at the same position removes the loudspeaker and the room exactly, and what is left is the correction the system actually applied, with nothing modelled and nothing fitted. On the left main that ratio follows the designed filter to 0.23 dB RMS over 200 Hz to 10 000 Hz for the linear-phase export and to 0.25 dB for the minimum-phase one; on the right main, to 0.30 dB and 0.34 dB. The two renderings share a designed magnitude by construction and, over that band, they land on it equally well, although their peak samples are 0.98 and 1.72 on the left and 1.42 and 2.10 on the right: a correction that boosts anywhere is not bounded by full scale, and every stage that stores it has to carry samples above it. The design of Section 6, the exported impulse response, the convolution engine and the routing therefore deliver in the room the filter that was asked for.

Below 200 Hz the subwoofer is part of the sum and that ratio is no longer the correction alone, so the crossover region is checked against the prediction of (9) instead, which Figure 1(b) does for the three measured states at once. Over 35 Hz to 250 Hz the model reproduces every one of the six cases, three states on two mains, to between 0.87 dB and 0.93 dB RMS, worst position 1.27 dB, and their level through the crossover band to 0.43 dB, across the 5 dB that separates the uncorrected state from the corrected ones. The same measurement confirms the latency bookkeeping of Section 3. The linear-phase filter peaks 1026 samples in, at 23.3 ms, yet the measured system delay is 47.2 ms to 48.1 ms in all three states alike rather than 23 ms more in the one carrying it, the filter's bulk latency having been absorbed into the 29.7 ms alignment delay the rig needed anyway.

The two exports are also the measured control on the alignment itself. They are rendered from one analysis, they are given the same output delay and the same subwoofer feed, and the minimum-phase rendering drops the designed phase and with it the all-pass (Section 7). Through the crossover band $[f_x/\sqrt{2}, f_x\sqrt{2}]$ the aligned export carries 0.47 dB more level than the minimum-phase one on the left main and 0.47 dB on the right, at every one of the ten positions on both, from 0.17 dB at the least to 0.82 dB at the most. The model puts that same difference at 0.32 dB and 0.31 dB.

That difference is not the all-pass alone, and the reason is worth stating because it bounds the experiment. At and above the crossover the two renderings carry the same designed magnitude, A having unit magnitude and the band-edge fade being complete there, but 2048 taps at 44.1 kHz do not deliver it: the two rendered magnitudes differ by 1.63 dB RMS over 85 Hz to 200 Hz on the left main, 4.1 dB at the worst frequency, and by 0.77 dB on the right, where over 200 Hz to 1000 Hz they agree to 0.12 dB and 0.30 dB and above 1 kHz to a hundredth of a decibel. A linear-phase filter spreads its response over ± 23 ms and a minimum-phase one front-loads it, and at 100 Hz neither length has room for the structure the design asks for, so the two fail differently. Holding the magnitude fixed and swapping only the phase separates the two effects: on the minimum-phase rendering's magnitude the aligned phase is worth 0.21 dB of band level on the left main and 0.42 dB on the right, and on the linear-phase rendering's magnitude 0.31 dB and 0.44 dB. Between a fifth and a half of a decibel is therefore the alignment, and the remainder of the measured 0.47 dB is two renderings disagreeing about a magnitude they were meant to share. The cure is length. Rendering the same design at 4096 taps brings the two magnitudes to within 0.29 dB to 0.45 dB RMS over the same band and at 8192 to 0.06 dB, at which point the experiment is a phase experiment; the price is that a linear-phase filter of 8192 taps carries 92.9 ms of bulk latency, more than the alignment delay can absorb, so the compensation of Section 3 adds the remaining 63 ms to every

output. The relative timing the alignment depends on is untouched by that, which is what makes the longer-filter repeat a valid experiment rather than a different one.

Figure 2 shows what the length does to the crossover region itself. The prediction at 2048 taps follows the measured state, dip for dip, including a 6 dB cancellation at 115 Hz on the left main that is not in the design and is gone by 4096 taps; from 8192 upwards the curves lie on top of one another, the filter having become long enough to stop being part of the answer. The dip is the clearest thing in the figure and it is an artefact of the rendering, which is worth knowing before reading anything else into a single frequency of a corrected response. What an alignment buys through a crossover is in any case not mainly level but coherence, which no single system measurement separates from the room; the level through the band is the part of it a measurement can hold, and it moves in the direction and by the order of magnitude the model gives.

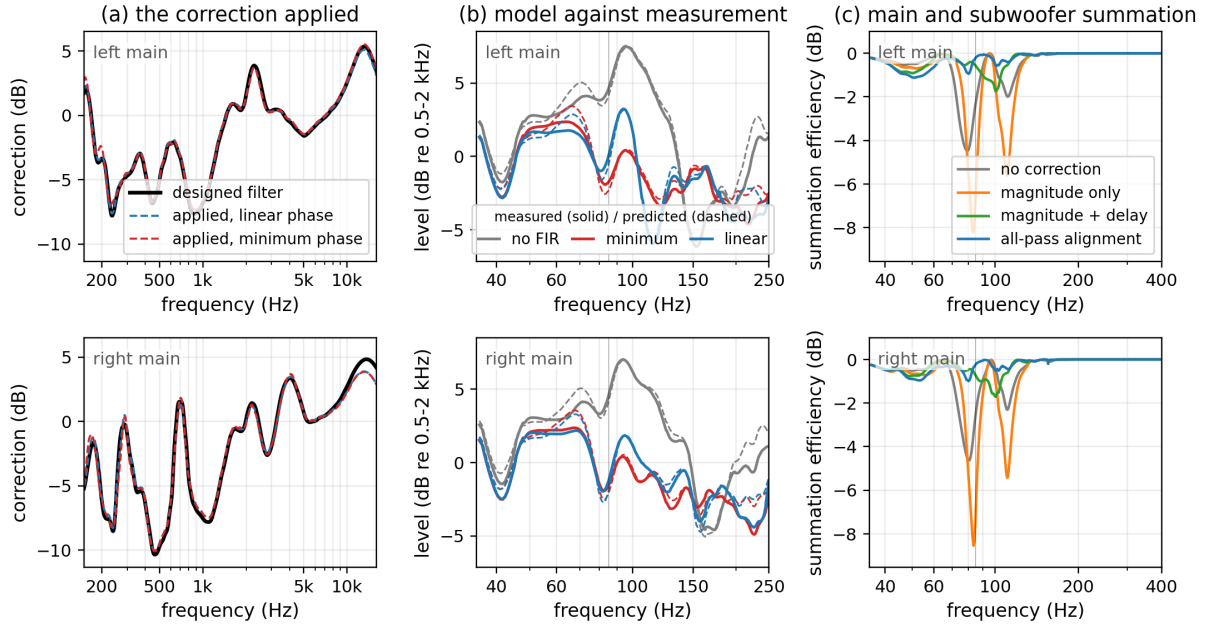


Figure 1: Both main loudspeakers with the subwoofer, the left on the top row and the right on the bottom; each column shares its vertical range between the two. (a) The correction the system applies, measured as the ratio of each corrected run to the uncorrected one at the same position, against the filter that was designed; the two phase renderings share that magnitude and both realise it. (b) The three measured states through the crossover, averaged over the ten positions, against the prediction of (9). The left main’s aligned state gains 1.15 dB of band level over the minimum-phase one across 85 Hz to 120 Hz and gives 0.33 dB of it back across 100 Hz to 140 Hz, where the two renderings disagree most about their common designed magnitude; the right main, whose renderings agree better, is ahead in both bands. (c) Summation efficiency through the crossover for four strategies, averaged over the ten positions. Zero is coherent addition.

Table 1 scores the strategies over $[f_x/2, 2f_x]$. Summation efficiency is $20 \log_{10} |M+S| - 20 \log_{10} (|M| + |S|)$ weighted by the overlap of the two sources, so 0 dB is perfect phase agreement and negative values are cancellation; $|\Delta\phi|$ is the overlap-weighted RMS phase difference between main and subwoofer, and the notch column the worst single-frequency cancellation.

Four things follow. Correcting the main’s magnitude on its own makes the summation worse than leaving the system alone, by 0.5 dB, because a minimum-phase magnitude correction moves the main’s phase through the crossover region without regard for the subwoofer. Most of what is then recovered comes from the bulk delay, and the all-pass takes the remainder, leaving -0.39 dB of efficiency and a worst cancellation of 1.7 dB where the magnitude-corrected system loses 10 dB and the untouched one 5.6 dB. The aligned system runs from -0.10 dB at the best of the ten positions to -0.94 dB at the worst, and it is the best of the five strategies at seven of them on the left main and six on the right; where it is beaten, it is beaten by the arrival-time delay, which on this rig is a close second at -0.45 dB. What separates them is not the average but the robustness, which the last two rows and the next paragraph are about. The leave-one-out scores match the in-sample ones to 0.02 dB, so the subwoofer’s phase through

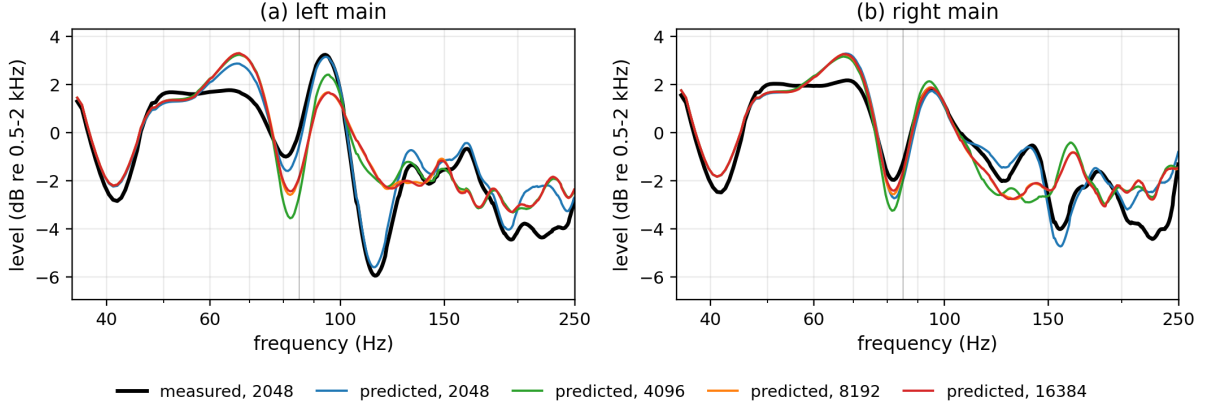


Figure 2: Filter length in the crossover region, for the two mains. Each curve is the linear-phase state averaged over the ten positions: the measured one, whose filter was rendered at 2048 taps, and the same design rendered at four lengths and put through (9). The 2048 prediction tracks the measurement; the longer renderings converge, and 8192 and 16384 overlay each other.

strategy	summation efficiency (dB)		$ \Delta\varphi $ (deg)	notch (dB)
	left	right		
no correction, no alignment	-1.27	-1.31	79	-5.6
magnitude correction only	-1.79	-1.81	82	-10.1
magnitude and arrival-time delay	-0.45	-0.45	42	-1.5
magnitude and crossover-band delay	-0.76	-0.77	52	-1.9
magnitude, crossover-band delay and all-pass	-0.39	-0.39	42	-1.7
as above, with the subwoofer inverted	-1.05	-0.84	75	-4.5
as above, without the bulk delay	-0.58	-0.59	47	-2.3

Table 1: Main-to-subwoofer summation over $[f_x/2, 2f_x]$, ten positions, designed leave-one-position-out.

the crossover behaves as a property of the room and the rig rather than of one seat, which is what justifies designing the alignment from a spatial average.

The all-pass absorbs a subwoofer polarity inversion, but only as far down as the analysis band edge lets it reach. The correction is faded back to unity over a half-octave skirt below that edge (Section 6), and the alignment factor fades with it, so an edge set at the crossover itself, as it was here, switches the steering off below 60 Hz — inside the region the subwoofer dominates. Inverting the subwoofer then costs the aligned system 0.66 dB on the left main and 0.45 dB on the right, against 5.3 dB for an alignment by delay alone. Moving the edge to 60 Hz, an octave below nothing but with the skirt now clear of the crossover, restores it: the inverted and non-inverted cases score -0.31 dB and -0.39 dB, within the 0.1 dB the earlier five-position set also showed, and the aligned score itself does not move. Setting the analysis range to start below the crossover rather than at it is therefore worth doing, and it costs nothing, since the correction there is faded out in either case.

Three limits of this evaluation should be stated. First, the two delay estimators of Section 7 disagree by 0.7 ms on this rig and the table does separate them, in favour of the arrival-time one, by 0.31 dB. The geometry says why that should not be read as a general result: it places the subwoofer between 1.45 m and 1.75 m from the ten positions, a spread of 29 cm, of which its impulse-response peak reports 8 cm. The arrival time of a source band-limited to the crossover region is repeatable without being accurate, and on an earlier rig whose subwoofer arrival wandered by 6.6 ms across positions it was the crossover-band estimate that held. Under the all-pass the choice matters less: dropping the bulk delay altogether costs the aligned system 0.19 dB, where merely changing which estimator supplies it costs a delay-only alignment 0.31 dB. Second, the comparison between strategies is predicted rather than measured, and while the measured pair above tests the one step the exports can isolate, the rest of the table rests on the offline chain. Rendered at the same 2048 taps and compared like with like, that chain reproduces the plugin’s own exported filters to between 0.02 dB and 0.13 dB RMS in every band from 60 Hz to 8 kHz,

for both renderings and both mains, which is as close as two independent implementations of the same arithmetic are likely to come. Third, one rig with one subwoofer in one room is one case, and the numbers here are not a claim about rooms in general.

An earlier attempt at the same measured pair failed for a reason worth recording, because it is the kind of defect a validation of this sort is good at finding. A minimum-phase filter concentrates its energy in its first samples, so its peak sample is large, and the minimum-phase exports here peak 4.7 dB and 6.4 dB above digital full scale, one of the linear-phase ones by 3.0 dB. The plugin stored its own state through an integer-coded audio container, which clamped those samples to full scale, and on reload preferred that stored copy to the file on disk. The correction in circuit then matched the clamped filter to 0.24 dB RMS and the designed one only to 2.1 dB, and the two renderings could not be compared because neither was the filter it claimed to be. The container now carries 32 bit floating-point samples, deflated, and the ratio of Figure 1(a) is what shows the repair.

The scripts that produce the table and the two figures are in the repository named in Section 10.

10 Implementation and availability

SuperMoTo is written in C++ and built with CMake against the JUCE framework⁵. Its reusable parts (the correction and measurement DSP, the plotting components, the preset system) live in a separate library, FxmeTools⁶, shared with the author’s other plugins, and the parts that need no JUCE at all are compiled and unit-tested on their own. Per-output FIR convolution uses the WDL non-uniformly partitioned convolution engine⁷ [16].

The engine holds no parameters of its own. It pulls them from the shared model whenever the model’s version counter changes, under a try-lock that defers the pull to the next block if the editor happens to be writing, so a GUI edit or a state restore can never stall the audio callback, and every update path is written to allocate nothing on the audio thread. Impulse-response files are read, resampled and installed on the message thread under a lock that the audio thread only ever takes with a try-lock, so a load in flight costs a few unconvolved blocks rather than a blocked callback. Meters and the spectrum analyser are fed by lock-free taps placed on each crosspoint and each output.

Continuous-integration builds for Linux, macOS (Apple Silicon universal binary) and Windows are provided by a GitHub Actions workflow triggered by version tags. The plugin is available in VST3, AU, and Standalone formats with a fixed 32-in/32-out discrete channel bus, which sidesteps the inconsistent per-host discrete-channel-count negotiation of JUCE’s AU and VST3 wrappers, the active matrix size being an independent, user-selectable setting up to that count. A user manual covering the workflow, the controls and the signal processing behind them is written in LaTeX alongside the source.

The source, the manual, the unit tests and the scripts that produce Table 1 and Figures 1 and 2 are at <https://github.com/odoare/SuperMoTo>, under the AGPL 3.0 or commercial terms; the manual, the paper and the scripts themselves are under the LGPL 3.0.

11 Conclusion

SuperMoTo puts the parts of a monitoring chain that are usually separate tools into one plugin instance, so that the routing, the per-loudspeaker correction, the time alignment and the Ambisonics decoding of a listening room are described in one place and calibrated without leaving it. The subwoofer alignment is the part that gains most from that arrangement. Because the measurement, the spatial average and the filter design live in the same instance, the alignment can be read straight off the measured phase in closed form and folded into the correction filter, where a tool working from outside the signal chain would have to fit a separate parametric all-pass to it.

What the approach cannot do is bounded by what any causal filter can do in a room. Cancellation notches and late reflections are not minimum-phase and are not inverted here, the correction is designed for the area the microphone positions span rather than for one seat, and the subwoofer alignment holds

⁵<https://juce.com/>

⁶<https://github.com/odoare/FxmeTools>

⁷<https://www.cockos.com/wdl/>

over that same area. Its effect on the summation through the crossover is quantified in Section 9, where it leaves the two sources within 0.4 dB of coherent addition across a ten-position listening area.

12 Acknowledgements

Generative AI (Claude, Anthropic) was used in the preparation of this work: to revise parts of the manuscript, to improve prior art, and to help writing the analysis scripts that produce the offline validation of Section 9. All measurements were made by the author, who checked the analysis against the plugin’s own implementation and against the measured system, verified the cited literature, and takes full responsibility for the content of the paper. Design, modelling, and DSP implementation of SuperMoTo was performed by the author. Threading, GUI, and project management was worked on with the help of the same Generative AI tools.

References

- [1] P. D. Welch, “The use of fast Fourier transform for the estimation of power spectra: a method based on time averaging over short, modified periodograms,” *IEEE Trans. Audio Electroacoust.*, vol. 15, no. 2, pp. 70–73, 1967.
- [2] A. Novak, P. Lotton and L. Simon, “Synchronized swept-sine: theory, application, and implementation,” *J. Audio Eng. Soc.*, vol. 63, no. 10, pp. 786–798, 2015.
- [3] J. N. Mourjopoulos, “Digital equalization of room acoustics,” *J. Audio Eng. Soc.*, vol. 42, no. 11, pp. 884–900, 1994.
- [4] S. T. Neely and J. B. Allen, “Invertibility of a room impulse response,” *J. Acoust. Soc. Am.*, vol. 66, no. 1, pp. 165–169, 1979.
- [5] P. D. Hatziantoniou and J. N. Mourjopoulos, “Generalized fractional-octave smoothing of audio and acoustic responses,” *J. Audio Eng. Soc.*, vol. 48, no. 4, pp. 259–280, 2000.
- [6] O. Kirkeby and P. A. Nelson, “Digital filter design for inversion problems in sound reproduction,” *J. Audio Eng. Soc.*, vol. 47, no. 7/8, pp. 583–595, 1999.
- [7] S. H. Linkwitz, “Active crossover networks for noncoincident drivers,” *J. Audio Eng. Soc.*, vol. 24, no. 1, pp. 2–8, 1976.
- [8] SMPTE ST 2095-1:2015, *Calibration Reference Wideband Digital Pink Noise Signal*. White Plains, NY: Society of Motion Picture and Television Engineers, 2015.
- [9] S. Bharitkar and C. Kyriakakis, “Phase equalization for multi-channel loudspeaker-room responses,” U.S. Patent 7 720 237 B2, 2010 (priority 7 Sept. 2004).
- [10] S. Bharitkar and C. Kyriakakis, *Immersive Audio Signal Processing*. New York: Springer, 2006.
- [11] L.-J. Brännmark, F. Rosencrantz and M. Ueno Andersson, *Dirac Live Bass Control*. Dirac Research, technical white paper, 2020.
- [12] J. Kim, “Group delay-driven crossover optimization for subwoofer satellite systems at listening position,” *Acta Acustica*, vol. 9, art. 51, 2025.
- [13] C. H. Knapp and G. C. Carter, “The generalized correlation method for estimation of time delay,” *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 24, no. 4, pp. 320–327, 1976.
- [14] J. Vanderkooy, “The acoustic center: a new concept for loudspeakers at low frequencies,” in *Proc. 121st AES Convention*, San Francisco, 2006.
- [15] F. E. Toole, *Sound Reproduction: The Acoustics and Psychoacoustics of Loudspeakers and Rooms*, 3rd ed. New York: Routledge, 2017.
- [16] W. G. Gardner, “Efficient convolution without input-output delay,” *J. Audio Eng. Soc.*, vol. 43, no. 3, pp. 127–136, 1995.
- [17] A. V. Oppenheim and R. W. Schaffer, *Discrete-Time Signal Processing*, 3rd ed. Upper Saddle River, NJ: Prentice Hall, 2010.

- [18] C. Nachbar, F. Zotter, E. Deleflie, and A. Sontacchi, “AmbiX — a suggested ambisonics format,” in *Proc. 3rd Ambisonics Symposium*, Lexington, 2011.
- [19] F. Zotter and M. Frank, *Ambisonics: A Practical 3D Audio Theory for Recording, Studio Production, Sound Reinforcement, and Virtual Reality*. Cham: Springer, 2019 (open access).